

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set plotting style for clean visualization
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (12, 8)
```

```
In [2]: np.random.seed(42)
n_samples = 1000

# Continuous Features
age = np.random.normal(38, 10, n_samples).astype(int)
income = np.random.exponential(scale=50000, size=n_samples) + 20000
credit_score = np.random.randint(300, 850, n_samples)

# Make Skill Wallet Balance heavily dependent on Income and Age
skill_balance = (income * 0.01) + (age * 5) + np.random.normal(200, 100, n_samples)
skill_balance = np.clip(skill_balance, 0, None) # No negative balances

# Categorical Features
employment = np.random.choice(['Employed', 'Self-Employed', 'Freelancer', 'Unemploy
education = np.random.choice(['High School', 'Bachelor', 'Master', 'PhD'], size=n_s

# Make Approval Status dependent on combinations of features (Multivariate Logic)
log_odds = (credit_score - 600) * 0.02 + (income / 30000) + (skill_balance / 200) -
probs = 1 / (1 + np.exp(-log_odds))
approval = np.where(probs > np.random.uniform(0, 1, n_samples), 1, 0)

df = pd.DataFrame({
    'Applicant_Age': age,
    'Annual_Income': income,
    'Credit_Score': credit_score,
    'Skill_Wallet_Balance': skill_balance,
    'Employment_Status': employment,
    'Education_Level': education,
    'Approval_Status': approval
})

print("Dataset successfully generated. Rows:", df.shape[0], "Columns:", df.shape[1])
print("\n")
```

Dataset successfully generated. Rows: 1000 Columns: 7

```
In [3]: print("--- Calculating Feature Correlation Matrix ---")
num_cols = ['Applicant_Age', 'Annual_Income', 'Credit_Score', 'Skill_Wallet_Balance']
corr_matrix = df[num_cols].corr()
print(corr_matrix.round(3))
print("\n")

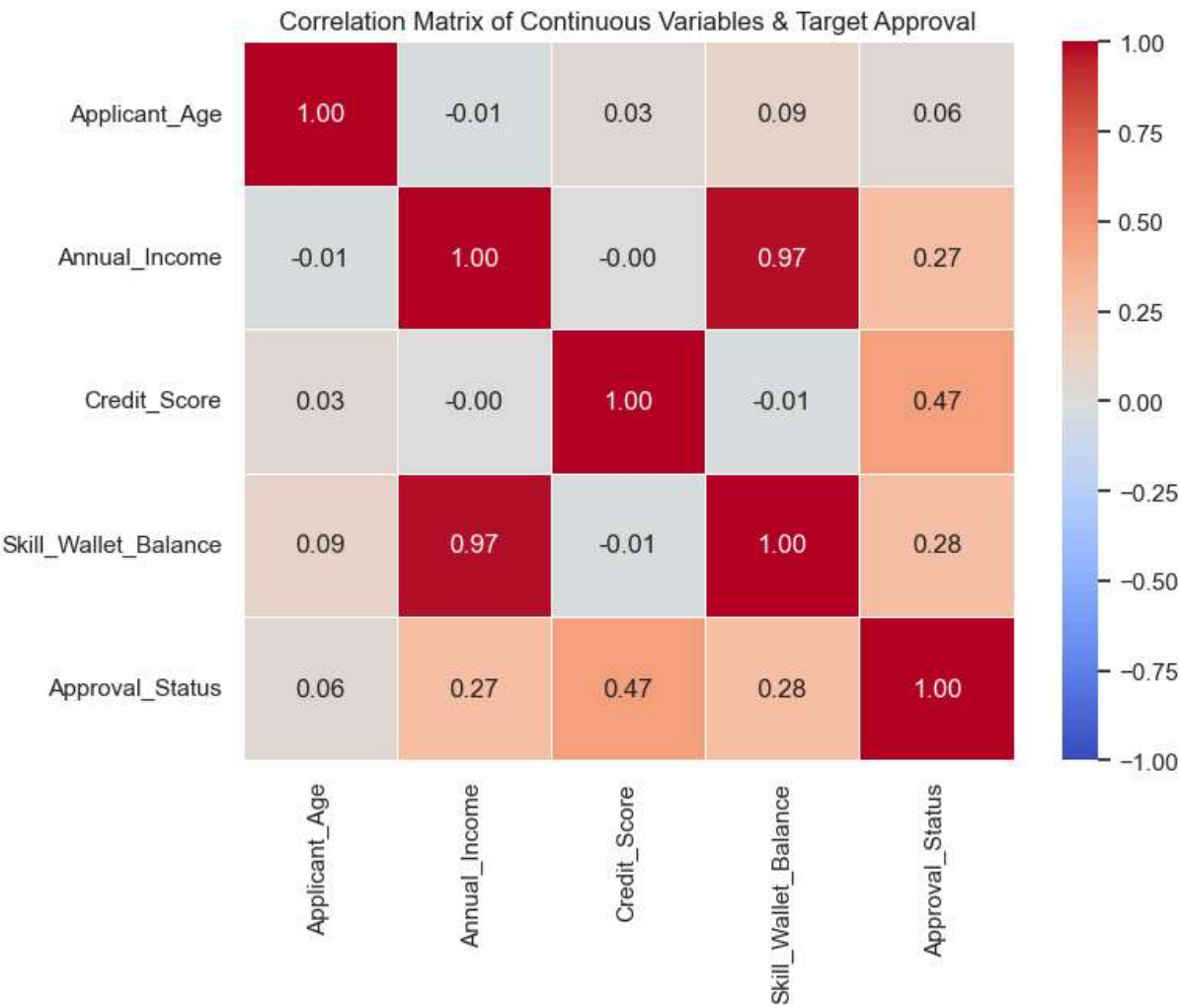
# Visualizing Correlations with a Heatmap
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plt.figure(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5, vm
plt.title('Correlation Matrix of Continuous Variables & Target Approval')
plt.show()
```

--- Calculating Feature Correlation Matrix ---

	Applicant_Age	Annual_Income	Credit_Score \
Applicant_Age	1.000	-0.010	0.035
Annual_Income	-0.010	1.000	-0.001
Credit_Score	0.035	-0.001	1.000
Skill_Wallet_Balance	0.094	0.974	-0.009
Approval_Status	0.057	0.267	0.466

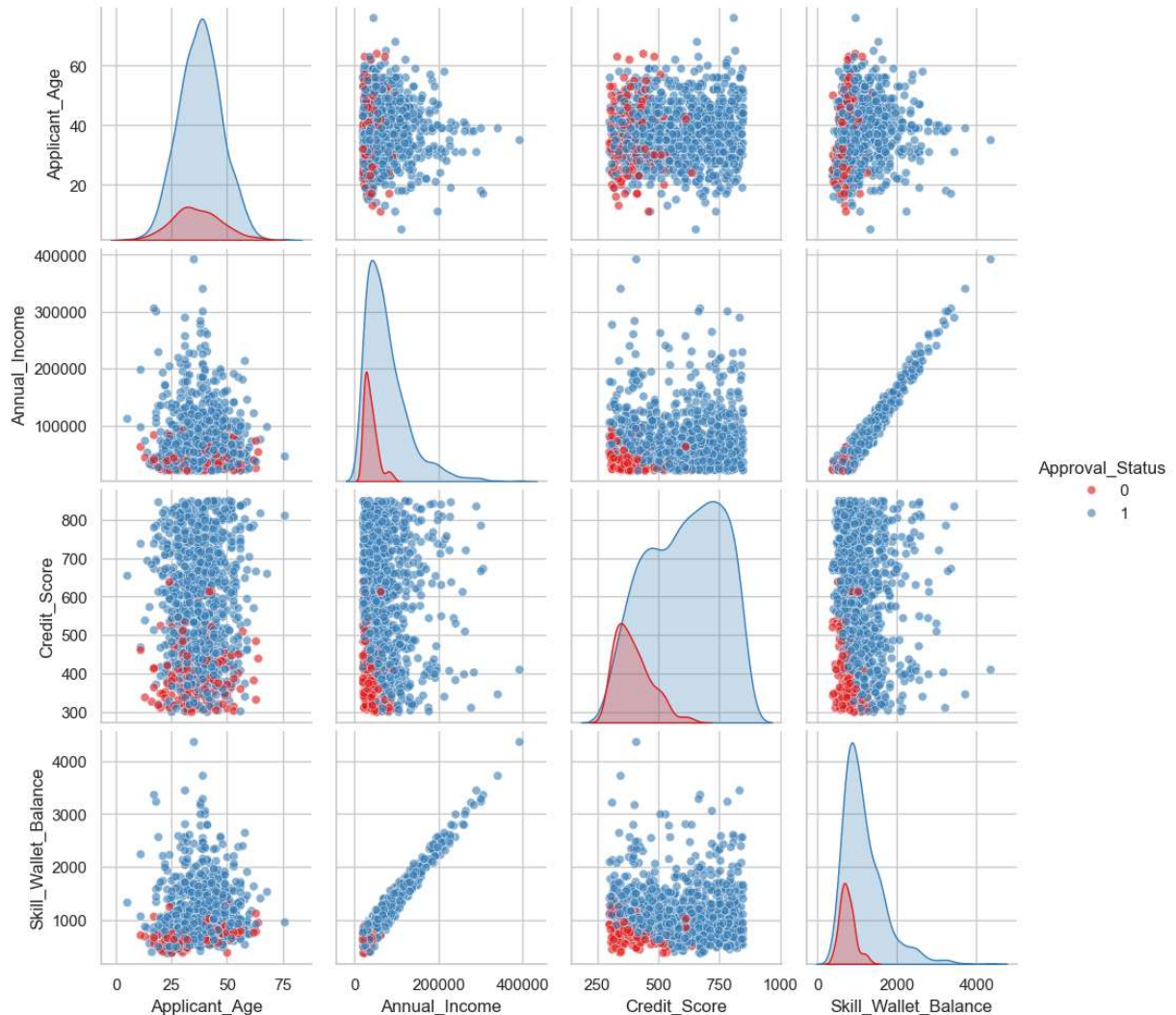
	Skill_Wallet_Balance	Approval_Status
Applicant_Age	0.094	0.057
Annual_Income	0.974	0.267
Credit_Score	-0.009	0.466
Skill_Wallet_Balance	1.000	0.284
Approval_Status	0.284	1.000



```
In [4]: print("--- Generating Multi-Variable Pairplot ---")
pair_plot = sns.pairplot(df[num_cols], hue='Approval_Status', palette='Set1', diag_
pair_plot.fig.suptitle('Pairwise Feature Interactions Segmented by Credit Card Appr
plt.show())
```

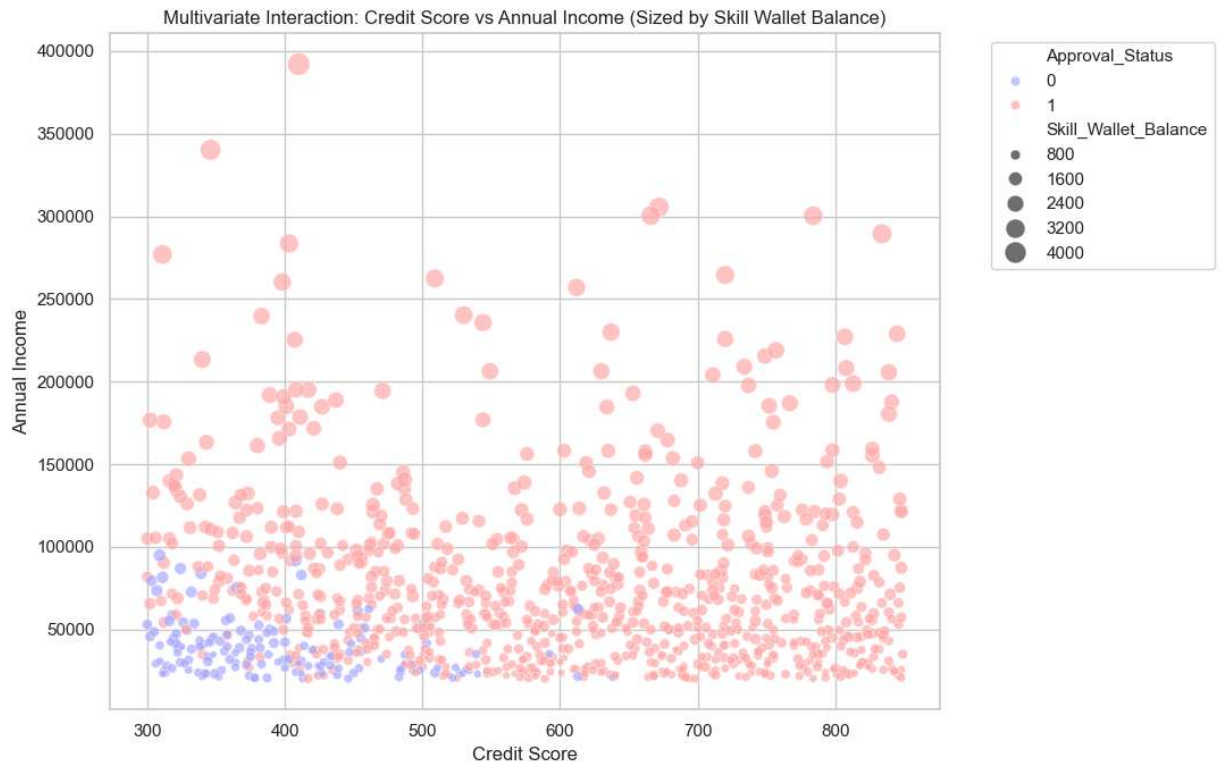
--- Generating Multi-Variable Pairplot ---

Pairwise Feature Interactions Segmented by Credit Card Approval Status

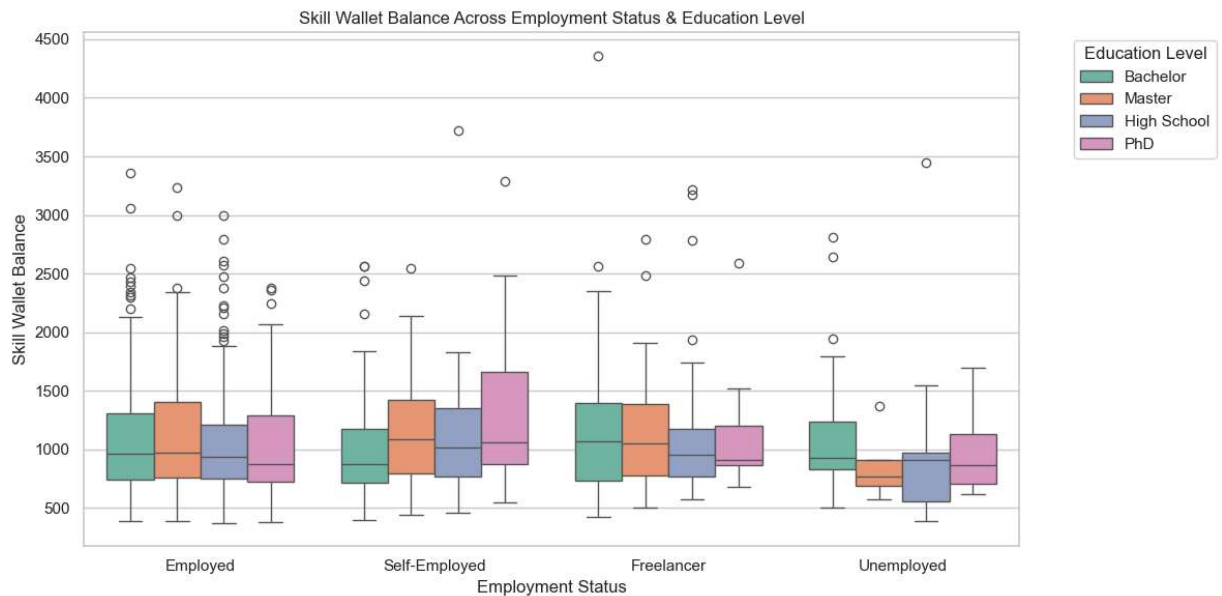


```
In [5]: plt.figure(figsize=(11, 7))
sns.scatterplot(
    data=df,
    x='Credit_Score',
    y='Annual_Income',
    hue='Approval_Status',
    size='Skill_Wallet_Balance',
    sizes=(20, 200),
    palette='bwr',
    alpha=0.7
)
plt.title('Multivariate Interaction: Credit Score vs Annual Income (Sized by Skill
plt.xlabel('Credit Score')
plt.ylabel('Annual Income')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
```

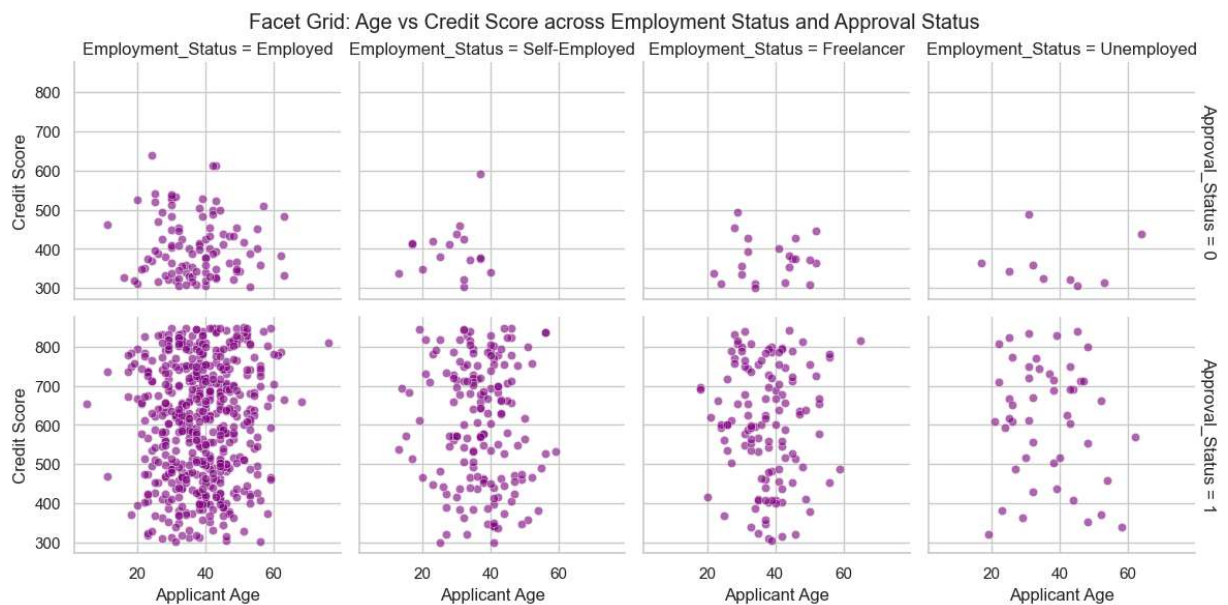
```
plt.tight_layout()
plt.show()
```



```
In [6]: plt.figure(figsize=(12, 6))
sns.boxplot(
    data=df,
    x='Employment_Status',
    y='Skill_Wallet_Balance',
    hue='Education_Level',
    palette='Set2'
)
plt.title('Skill Wallet Balance Across Employment Status & Education Level')
plt.xlabel('Employment Status')
plt.ylabel('Skill Wallet Balance')
plt.legend(title='Education Level', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



```
In [7]: g = sns.FacetGrid(df, col="Employment_Status", row="Approval_Status", margin_titles
g.map(sns.scatterplot, "Applicant_Age", "Credit_Score", color="purple", alpha=0.6)
g.set_axis_labels("Applicant Age", "Credit Score")
g.fig.subplots_adjust(top=0.9)
g.fig.suptitle('Facet Grid: Age vs Credit Score across Employment Status and Approv
plt.show()
```



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In [ ]:
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